

LECTURE NOTES MATH255

SVEN SANDFELDT

1. LECTURE 12

In this class we introduce vector spaces over a field F , and investigate how they relate to field extensions.

1.1. Vector spaces. Let F be a field. We say that a set V is a vector space over the field F (or a F -vector space) if V has an commutative binary operation $+$ which makes $(V, +)$ into an abelian group, and if there is a scalar multiplication $\cdot : F \times V \rightarrow V$, written av where $a \in F$ and $v \in V$, that satisfies:

- (i) $av \in V$ for all $a \in F$ and $v \in V$,
- (ii) $a(bv) = (ab)v$ for all $a, b \in F$ and $v \in V$,
- (iii) $(a + b)v = av + bv$ for all $a, b \in F$ and $v \in V$,
- (iv) $a(v + w) = av + aw$ for all $a \in F$ and $v, w \in V$,
- (v) $1v = v$ where 1 is the multiplicative identity in F and for all $v \in V$.

We usually call the elements $v \in V$ *vectors* and the elements of F *scalars*.

Example 1.1. Let $\mathbb{R}^n = \mathbb{R} \times \dots \times \mathbb{R}$, then we define addition in $\mathbb{R}^n$ as $(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n)$ and scalar multiplication as:

$$(1.1) \quad t(x_1, \dots, x_n) = (tx_1, \dots, tx_n).$$

With these operations the set $\mathbb{R}^n$ is a vector space over $\mathbb{R}$.

Example 1.2. We can generalize the previous example slightly. If F is any field, then $F^n = F \times \dots \times F$ is the product of n copies of F . We can define addition in F^n as:

$$(1.2) \quad (x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n),$$

and scalar multiplication as:

$$(1.3) \quad a(x_1, \dots, x_n) = (ax_1, \dots, ax_n), \quad a \in F, (x_1, \dots, x_n) \in F^n$$

and with these operations F^n is a vector space over F .

Example 1.3. Let $F \leq E$ be a field extension of a field F . Then we can view E as a vector space over F where addition is imply addition in E , and multiplication with scalars is just given by multiplying elements in E with elements in F (recalling that F is a subfield of E).

1.2. Spanning sets, linear independence, and dimension. Recall that if V is a vector space over F and $v_1, \dots, v_m \in V$ then a linear combination of $v_1, \dots, v_m$ is any vector of the form:

$$(1.4) \quad a_1 v_1 + \dots + a_m v_m \in V.$$

Definition 1.1. Let V be a vector space over F . A subset $S \subset V$ is *spanning*, or *span* V , if any element $v \in V$ can be written as a linear combination of the elements in S .

Definition 1.2. Let V be a vector space over F . A set $S \subset V$ is *linearly independent* if for any $v_1, \dots, v_n \in S$ and $a_1, \dots, a_n \in F$ such that:

$$(1.5) \quad a_1 v_1 + \dots + a_n v_n = 0$$

we have $a_1 = a_2 = \dots = a_n = 0$.

Definition 1.3. Let V be a vector space over F . The vector space V is *finite dimensional* if there is a finite set $S \subset V$ that spans V . The *dimension* of V is ∞ if V is not finite dimensional, if V is finite dimensional then the dimension of V is n if n is the minimal number of elements in a set $S \subset V$ that spans V .

Definition 1.4. Let V be a vector space over F . A *basis* of V is a subset $S \subset V$ that is both linearly independent and span V .

Remark 1. As you will have seen in linear algebra class, the number of elements in any basis of a vector space is equal to the dimension of the vector space. In particular, if $\dim(V) = n < \infty$ then any basis contains n elements. In fact, this is an equivalent definition of dimension.

Theorem 1.1. Let V be a vector space over F . If $\dim(V) = n$ and $v_1, \dots, v_m \in V$ are elements where $m > n$ then there is some $a_1, \dots, a_m \in F$ not all zero such that:

$$(1.6) \quad a_1 v_1 + \dots + a_m v_m = 0.$$

Proof. The only way that the theorem fails is if the elements $v_1, \dots, v_m$ are linearly independent. If the elements $v_1, \dots, v_m$ are linearly independent then we define $W = \{a_1 v_1 + \dots + a_m v_m \mid a_1, \dots, a_m \in F\}$ which is a subspace of V . Notice that the dimension of W has to be at most n since V has dimension n , so we conclude that any basis of W has at most n elements. But, $v_1, \dots, v_m$ is a basis of W which contains $m > n$ elements, this is a contradiction. We conclude that the elements $v_1, \dots, v_m$ can not be linearly independent and therefore that there must exist $a_1, \dots, a_m$ not all zero such that:

$$(1.7) \quad a_1 v_1 + \dots + a_m v_m = 0,$$

and this finishes the proof. ■

1.3. Extension fields and vector spaces. We have seen that if F is a field and $F \leq E$ is an extension field of F , then E is a vector space over F . In this last section of the notes we will apply ideas of vector spaces to prove statements about extension fields. However, we will do this for a special type of extension fields which are called *simple extensions*.

Definition 1.5. Let F be a field and $F \leq E$ a field extension. Let $\alpha \in E$ be algebraic over F . We define the field $F(\alpha)$ as the smallest subfield of E that contains both F and α . The field $F(\alpha)$ is said to be a simple extension of F .

Theorem 1.2. Let F be a field, $F \leq E$ a field extension and $\alpha \in E$ algebraic over F with degree n . Then we can write any element in $x \in F(\alpha)$ uniquely as:

$$(1.8) \quad x = a_0 + a_1\alpha + \dots + a_{n-1}\alpha^{n-1}, \quad a_0, \dots, a_{n-1} \in F.$$

Equivalently, $1, \alpha, \alpha^2, \dots, \alpha^{n-1}$ is a basis of $F(\alpha)$ when $F(\alpha)$ is viewed as a vector space over F . In fact, the field $F(\alpha)$ is isomorphic to $F[x]/\langle f_\alpha(x) \rangle$ where $f_\alpha(x) \in F[x]$ is the irreducible polynomial such that $f_\alpha(\alpha) = 0$.

Proof. Let $f_\alpha(x) = a_0 + \dots + a_n x^n$ be the irreducible polynomial for α . Note that $1, \alpha, \dots, \alpha^j, \dots \in F(\alpha)$ and any linear combination of these elements are in $F(\alpha)$ since $\alpha \in F(\alpha)$, $F \subset F(\alpha)$ and $F(\alpha)$ is a field. We define a map $\psi : F[x]/\langle f_\alpha(x) \rangle \rightarrow F(\alpha)$ by:

$$(1.9) \quad \psi(p(x) + \langle f_\alpha(x) \rangle) = p(\alpha).$$

To see that ψ is well-defined, note that if $p(x) + \langle f_\alpha(x) \rangle = q(x) + \langle f_\alpha(x) \rangle$ then $p(x) - q(x) = g(x)f_\alpha(x) \in \langle f_\alpha(x) \rangle$ so we have:

$$(1.10) \quad p(\alpha) = q(\alpha) + g(\alpha)f_\alpha(\alpha) = q(\alpha)$$

where we have used that $f_\alpha(\alpha) = 0$. Moreover, notice that ψ is a homomorphism of fields since:

$$\begin{aligned} \psi(p(x) + q(x) + \langle f_\alpha(x) \rangle) &= p(\alpha) + q(\alpha) = \psi(p(x) + \langle f_\alpha(x) \rangle) + \psi(q(x) + \langle f_\alpha(x) \rangle), \\ \psi(p(x)q(x) + \langle f_\alpha(x) \rangle) &= p(\alpha)q(\alpha) = \psi(p(x) + \langle f_\alpha(x) \rangle)\psi(q(x) + \langle f_\alpha(x) \rangle), \\ \psi(1 + \langle f_\alpha(x) \rangle) &= 1. \end{aligned}$$

The inclusion $F \subset \text{Im}(\psi)$ follows since for any $a \in F$ we have $\psi(a + \langle f_\alpha(x) \rangle) = a \in F$. Since any field homomorphism is injective (try to prove this! It follows since any homomorphism between fields maps unity to unity, this can be used to show that the kernel is trivial), this proves that $\text{Im}(\psi) = F(\alpha)$ since $\text{Im}(\psi)$ is a subfield of $F(\alpha)$, the image of ψ contains F , and $\psi(x + \langle f_\alpha(x) \rangle) = \alpha$ so $\alpha \in \text{Im}(\psi)$. So, $\text{Im}(\psi)$ is a subfield of $F(\alpha)$ that contains F and α , but $F(\alpha)$ is the smallest field that contains F and α so we must have $\text{Im}(\psi) = F(\alpha)$. To get the final decomposition, we claim that any:

$$(1.11) \quad p(x) + \langle f_\alpha(x) \rangle \in F[x]/\langle f_\alpha(x) \rangle$$

can be represented by a polynomial $r(x)$ of degree at most $n - 1$, $p(x) + \langle f_\alpha(x) \rangle = r(x) + \langle f_\alpha(x) \rangle$. This is a consequence of the division algorithm: if $p(x) \in F[x]$ is arbitrary then we can write $p(x) = q(x)f_\alpha(x) + r(x)$ with $\deg(r(x)) < \deg(f_\alpha(x)) = n$. Since $q(x)f_\alpha(x) \in \langle f_\alpha(x) \rangle$ we have $q(x)f_\alpha(x) + \langle f_\alpha(x) \rangle = 0 + \langle f_\alpha(x) \rangle$ from which it follows that $p(x) + \langle f_\alpha(x) \rangle = r(x) + \langle f_\alpha(x) \rangle$. It follows that any element of $F[x]/\langle f_\alpha(x) \rangle$ can be written:

$$(1.12) \quad b_0 + b_1x + \dots + b_{n-1}x^{n-1} + \langle f_\alpha(x) \rangle,$$

and if we apply ψ and note that ψ is an isomorphism then we see that any element of $F(\alpha)$ can be written as:

$$(1.13) \quad \psi(b_0 + b_1x + \dots + b_{n-1}x^{n-1} + \langle f_\alpha(x) \rangle) = b_0 + b_1\alpha + \dots + b_{n-1}\alpha^{n-1}.$$

The last claim is that $1, \alpha, \dots, \alpha^{n-1} \in F(\alpha)$ are linearly independent (note that they span $F(\alpha)$). One way to see this is to note that if $b_0, \dots, b_{n-1} \in F$ are such that:

$$(1.14) \quad b_0 + b_1\alpha + \dots + b_{n-1}\alpha^{n-1} = 0$$

then α is a zero of the polynomial $g(x) = b_0 + b_1x + \dots + b_{n-1}x^{n-1} \in F[x]$. But if α is a zero of $g(x) \in F[x]$ then $f_\alpha(x)$ divides $g(x)$, so $g(x) = q(x)f_\alpha(x)$. Noting that $g(x)$ has degree at most $n-1$, and that $q(x)f_\alpha(x)$ has degree at least n , unless $q(x)$ is zero, we conclude that $q(x)$ must be zero. Since $q(x) = 0$ we obtain $g(x) = q(x)f_\alpha(x) = 0$ which implies $b_0 = b_1 = \dots = b_{n-1} = 0$. That is, if $b_0 + b_1\alpha + \dots + b_{n-1}\alpha^{n-1} = 0$ then $b_0 = \dots = b_{n-1} = 0$, so $1, \alpha, \dots, \alpha^{n-1} \in F(\alpha)$ are linearly independent over F , and since they also span $F(\alpha)$ we conclude that they form a basis of $F(\alpha)$. ■

Example 1.4. To see an example of the previous theorem, we can let $F = \mathbb{Q}$, $E = \mathbb{C}$ and $\alpha = \sqrt{7}$. In that case we have $\mathbb{Q}(\sqrt{7}) = \{a + b\sqrt{7} \mid a, b \in \mathbb{Q}\}$ (note that we have previously written $\mathbb{Q}[\sqrt{7}]$ for this field, but in the remainder we will simply write $\mathbb{Q}(\sqrt{7})$ to stay consistent with the book). Another non-trivial fact that the previous theorem shows is that $\mathbb{Q}[x]/\langle x^2 - 7 \rangle$ is isomorphic to $\mathbb{Q}(\sqrt{7})$ (notice that the irreducible polynomial for $\sqrt{7}$ is $x^2 - 7$).

Example 1.5. The theorem above can be used to sometimes give more concrete description of quotient fields $F[x]/\langle f(x) \rangle$ when $f(x)$ is irreducible. If $f(x) \in \mathbb{Q}[x]$ is irreducible and α is a zero of $f(x)$, then $f(x)$ is the irreducible polynomial for α . The theorem above then shows that $\mathbb{Q}[x]/\langle f(x) \rangle$ is isomorphic to:

$$(1.15) \quad \mathbb{Q}(\alpha) = \{b_0 + b_1\alpha + \dots + b_{n-1}\alpha^{n-1} \mid b_0, \dots, b_{n-1} \in \mathbb{Q}\}$$

where n is the degree of $f(x)$.

Theorem 1.3. *Let F be a field $F \leq E$ a field extension and $\alpha \in E$ algebraic over F . Then $\dim(F(\alpha)) = \deg(\alpha)$.*

Proof. Note that $1, \dots, \alpha^{n-1}$ is a basis of $F(\alpha)$ as a vector space over F , where n is the degree of $f_\alpha(x)$ and therefore the degree of α . ■

Theorem 1.4. *Let F be a field and $F \leq E$ a field extension and $\alpha \in E$ algebraic over F . Then every element $\beta \in F(\alpha)$ is algebraic and is the zero of a polynomial of degree at most $n = \deg(\alpha)$.*

Proof. Note that $1, \beta, \dots, \beta^n \in F(\alpha)$ are not linearly independent since the dimension of $F(\alpha)$ is $\deg(\alpha)$. It follows that there are $b_0, \dots, b_n \in F$ not all zero such that:

$$(1.16) \quad 0 = b_0 + b_1\beta + \dots + b_n\beta^n$$

so β is a zero of the polynomial $g(x) = b_0 + \dots + b_nx^n$ which is a non-zero polynomial (since at least one $b_j \neq 0$) of degree at most n (note that the degree might not be n since we might have $b_n = 0$). ■

Email address: `sandfeldt@uchicago.edu`